```python
# 1. Install required libraries
!pip install -q transformers prophet seaborn

# 2. Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import re
from transformers import pipeline
from prophet import Prophet

# 3. Load dataset
df = pd.read_csv("/content/Corona_NLP_test.csv", encoding='latin1')
df = df[['TweetAt', 'OriginalTweet', 'Sentiment']]
df['TweetAt'] = pd.to_datetime(df['TweetAt'], dayfirst=True)

# 4. Clean tweet text
def clean_tweet(text):
    text = text.lower()
    text = re.sub(r"http\S+|@\S+|#\S+", "", text)  # Remove URLs, 
mentions, hashtags
    text = re.sub(r"[^a-z\s]", "", text)           # Keep only 
alphabets and spaces
    text = re.sub(r"\s+", " ", text)               # Remove extra 
spaces
    return text.strip()

df['CleanTweet'] = df['OriginalTweet'].apply(clean_tweet)

# 5. Sentiment Analysis using BERT
classifier = pipeline("sentiment-analysis", model="nlptown/bert-base-
multilingual-uncased-sentiment")

# Analyze only the first 200 tweets for demo purposes
sample_df = df.head(200).copy()
sample_df['PredictedSentiment'] = sample_df['CleanTweet'].apply(lambda 
x: classifier(x[:512])[0]['label'])

# 6. Simplify sentiment labels
def simplify(label):
    if "1" in label or "2" in label:
        return "Negative"
    elif "3" in label:
        return "Neutral"
    else:
        return "Positive"
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sample_df['PredictedSentiment'] =
sample_df['PredictedSentiment'].apply(simplify)

# 7. Plot original vs predicted sentiment
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
sns.countplot(data=sample_df, x='Sentiment')
plt.title("Original Sentiment")

plt.subplot(1, 2, 2)
sns.countplot(data=sample_df, x='PredictedSentiment')
plt.title("Predicted Sentiment (BERT)")
plt.tight_layout()
plt.show()

# 8. Time Series Aggregation of Sentiment
time_sentiment = sample_df.groupby(['TweetAt',
'PredictedSentiment']).size().reset_index(name='Count')
pivot_df = time_sentiment.pivot(index='TweetAt',
columns='PredictedSentiment', values='Count').fillna(0)

# Plot sentiment trend over time
pivot_df.plot(marker='o', figsize=(12, 6))
plt.title("Predicted Sentiment Over Time")
plt.ylabel("Tweet Count")
plt.grid(True)
plt.show()

# 9. Forecast Positive Tweets using Prophet
positive_ts =
pivot_df[['Positive']].reset_index().rename(columns={'TweetAt': 'ds',
'Positive': 'y'})

model = Prophet()
model.fit(positive_ts)

future = model.make_future_dataframe(periods=7)
forecast = model.predict(future)

# Plot forecast results
model.plot(forecast)
plt.title("Forecast of Positive Tweets")
plt.xlabel("Date")
plt.ylabel("Tweet Volume")
plt.show()

# Plot forecast components
model.plot_components(forecast)
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plt.show()
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